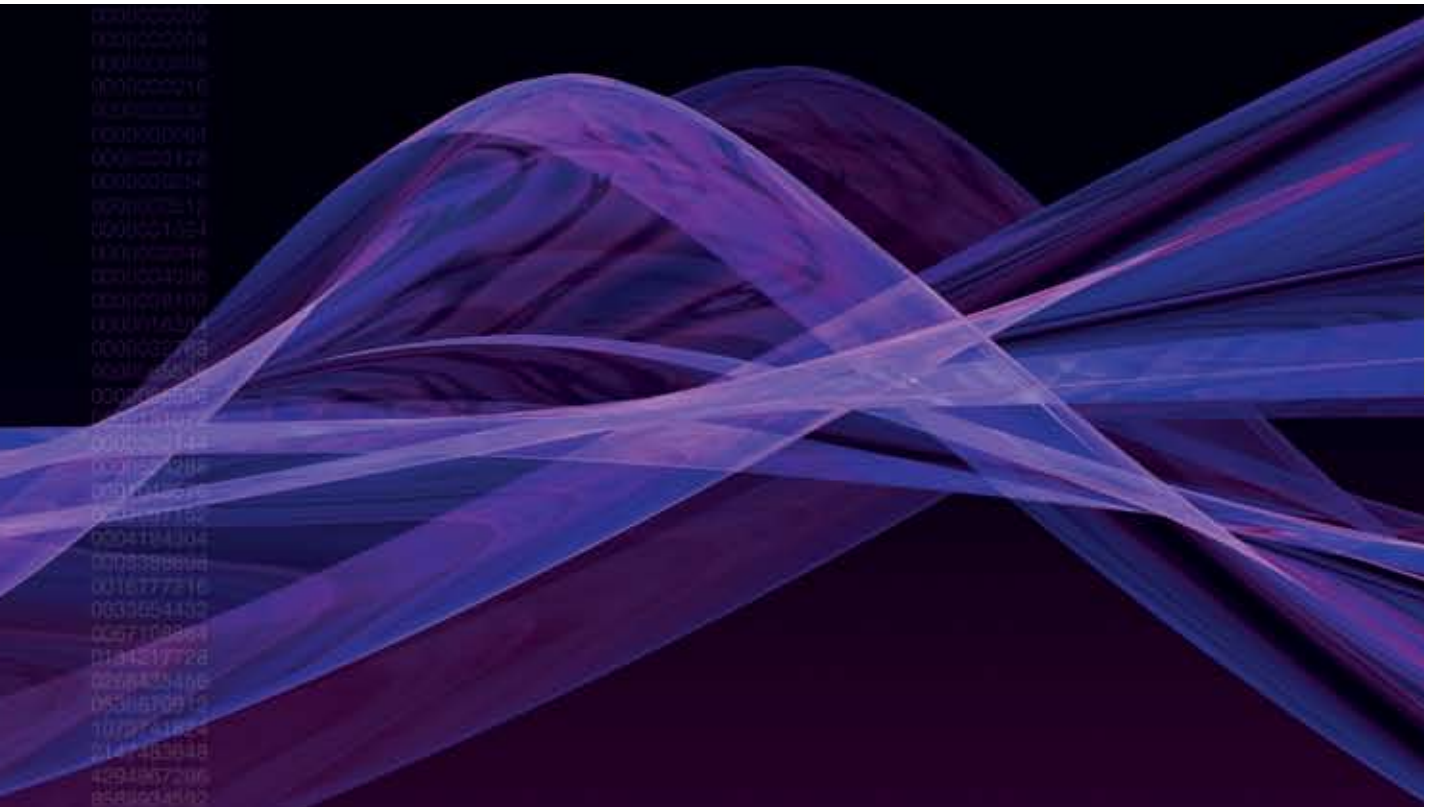




## Real-Time PCR

Amplification Instruments

**BIO-RAD**

## One Just Right for You

Real-time PCR detection is a powerful tool that simplifies DNA quantitation, genotyping, expression analysis, and many other applications. Bio-Rad's real-time PCR systems are designed to bring outstanding thermal performance and sensitive optical detection to all of your real-time PCR applications. By pairing proven real-time instruments with quality PCR reagents and powerful analysis software, you can consistently obtain high-quality real-time PCR data to fuel your research discoveries.

Bio-Rad's classic line of real-time PCR instruments presents valuable solutions for different research needs. Use the table on the following page to select the best instrument for you based on the features that are most important to your laboratory.



## Real-Time PCR Systems



**MiniOpticon™  
CFB-3120**



**MyiQ™  
170-9770**



**Chromo4™  
CFB-3240G**



**iQ™5  
170-9780**

|                                       |                         |            |                         |            |
|---------------------------------------|-------------------------|------------|-------------------------|------------|
| Number of targets detectable per well | 2                       | 1          | 4                       | 5          |
| Excitation range                      | 470–500 nm              | 475–495 nm | 450–650 nm              | 475–645 nm |
| Detection range                       | 523–700 nm              | 515–545 nm | 515–730 nm              | 515–700 nm |
| Gradient capability                   | •                       | •          | •                       | •          |
| Thermal gradient differential range   | 1–16°C                  | 1–25°C     | 1–24°C                  | 1–25°C     |
| Number of wells                       | 48                      | 96         | 96                      | 96         |
| Serial connectivity options           | Up to 4 units from 1 PC | —          | Up to 4 units from 1 PC | —          |
| Allelic discrimination                | •                       | —          | •                       | •          |
| End-point detection                   | —                       | •          | —                       | •          |
| Melt-curve analysis                   | •                       | •          | •                       | •          |
| Gene expression analysis              | •                       | •          | •                       | •          |

(•) Included; (—) Not available.

## All real-time systems from Bio-Rad provide:

- Unique modular design coupled with high-resolution optics for maximum flexibility and sensitivity
- Optical detection methods that enable a choice of real-time PCR chemistry
- Outstanding thermal performance from the DNA Engine®, MJ Mini™, or iCycler® thermal cycler
- Gradient functionality for rapid assay optimization
- Easy-to-use software for efficient, accurate data analysis
- Compatibility with outstanding PCR supermixes that provide optimum performance in real-time PCR assays



iQ5 System

MyiQ System

MiniOpticon System

Chromo4 System



The high-performance iQ5 real-time PCR detection system offers the ideal balance of power and flexibility for multiplex applications.

- The iQ5 system matches sensitive, filter-based optics with the gradient-capable 96-well block of the iCycler thermal cycler
- A high-resolution charge-coupled device (CCD)-based optical design optimizes the iQ5 system for multiplex analysis, enabling extremely reproducible quantitation of up to 5 targets simultaneously
- Data collected from the iQ5 system demonstrate outstanding uniformity over a broad dynamic range of input sample concentrations



The laboratory-friendly MyiQ system is the ideal real-time PCR workhorse for single-color real-time PCR applications.

- The MyiQ system brings straightforward single-color real-time PCR detection capabilities to the gradient-capable 96-well block of the iCycler thermal cycler
- With a CCD-based optical design optimized for detection of dyes such as SYBR Green I, FAM, VIC, TET, or JOE, the MyiQ system is well suited for quantitation and screening applications
- Option to upgrade to the MyiQ or iQ5 system from your existing iCycler thermal cycler



The compact MiniOpticon system places the most advanced real-time PCR applications and gel-free PCR analysis techniques well within reach.

- Based on the MJ Mini thermal cycler, the MiniOpticon system boasts precise thermal control and a powerful thermal gradient feature
- The MiniOpticon system incorporates a fixed array of 48 LEDs into an optical design with no moving parts, enabling reproducible and accurate multiplexing capability of up to 2 targets
- The MiniOpticon system is one of the smallest and most portable systems available for real-time PCR detection



The easy-to-use Chromo4 system delivers sensitive multiplex performance in a compact, 96-well design.

- The Chromo4 detector is an interchangeable Alpha™ unit that brings real-time quantitative PCR to the high-performance DNA Engine platform
- All components for fluorophore excitation and detection are brought directly to each well by the advanced optical design of the photonics shuttle
- Together with a DNA Engine or Dyad Disciple™ thermal cycler, the Chromo4 detector generates accurate multiplex data for up to 4 targets at once
- Option to upgrade to real-time PCR from your existing DNA Engine thermal cycler



## iQ5 Optical System Software for MyiQ and iQ5 Systems

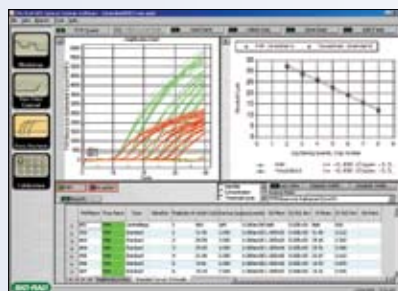
iQ5 optical system software, version 2.0, controls and analyzes data generated from the iQ5 and redesigned MyiQ real-time systems. The software can also analyze data files generated from earlier versions of the iCycler thermal cycler-based real-time family.

### Unmatched Flexibility

- Account for differences in reaction efficiency, and use multiple reference genes with advanced gene expression analysis methods
- Examine gene expression of multiple genes from the same plate easily using SYBR Green or probe-based detection
- Compare gene expression results from up to 5,000  $C_T$  data points and multiple different plates using the Gene Study module
- Identify genotyping samples automatically with flexible allelic discrimination analysis tool
- Simplify qualitative assessment of sample abundance in single- or multicolor assays with embedded tool for end-point fluorescence analysis

### Powerful Simplicity

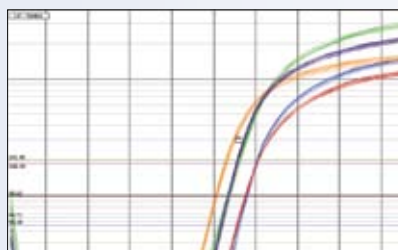
- Generate publication-quality statistics and graphs, quickly create customizable reports, or export data from tables to Microsoft Excel with 1-step export option
- Streamline your run setup and analysis by setting personal user preferences (such as volume, consumable, fluorophore, and analysis settings) using the user log-in feature, with optional password protection



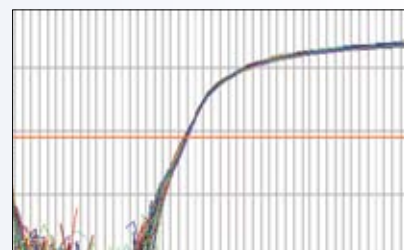
iQ optical system software displays amplification data and standard curves from all active fluorophores in a multiplex data set.



**Multifile gene expression analysis.**  
Compare  $C_T$  values from different data files with the Gene Study module.



**Five-target multiplex assay with FAM, TET, TAMRA, Texas Red, and Cy5.**



**96-well uniformity assay performed on the MyiQ system,  $SD = \pm 0.06 C_T$ .**

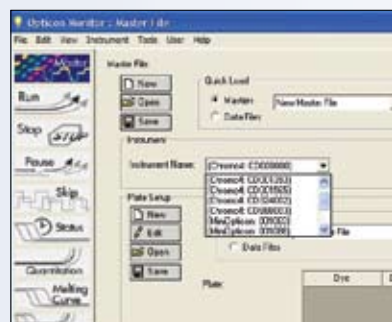
## iQ5 Optical System Software, Security Edition

- Offers all standard edition features, plus important U.S. FDA 21 CFR Part 11 compliance tools
- Works with Windows XP operating system security features to securely maintain, verify, and track electronic records generated by the software
- Features password-protected log-in, audit trail, file encryption, and electronic signature

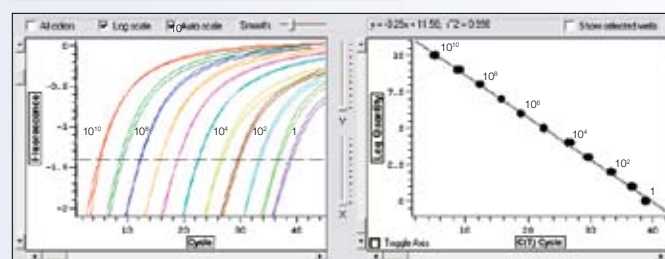
## Opticon Monitor™ Software for Chromo4 and MiniOpticon Systems

Go from start to run in just three steps, observe data acquisition in real time, and begin data analysis while the run is in progress.

- Collect data from all wells and channels every run
- Edit plate information before, during, or after a run
- Employ mouseover well highlighting feature
- Use Well Groups feature to analyze multiple experiments from a single plate
- Choose method for subtracting background signal and positioning threshold line
- Run any combination of up to four Chromo4 or MiniOpticon instruments from the same computer



Select connected instruments from the drop-down menu.



**Broad linear dynamic range with single-copy detection.**

## Genomic Research Solutions From Bio-Rad

Sample quantitation by real-time PCR detection is a key technique in genomics research. In addition to outstanding instruments for real-time PCR detection, Bio-Rad offers accessible, advanced technology for a variety of genomic research applications, from assay design to alternative gene expression analysis techniques.

### Real-Time PCR Assay Design

Beacon Designer probe and primer design software is an easy-to-use, comprehensive tool for real-time PCR assay design. This software package from PREMIER Biosoft International makes the design and selection of real-time PCR primers and probes effortless, even for applications such as multiplex gene expression analysis and allelic discrimination.

### Sample Preparation

Bio-Rad's nucleic acid isolation kits are a complete, extensive line of products for isolating and purifying DNA and RNA. From the purification of milligram quantities of plasmid DNA for transfection applications to rapid and efficient isolation of RNA for reverse transcription and real-time PCR or microarray applications, the Quantum Prep® plasmid kits and the Aurum™ RNA isolation kits are engineered to yield fast results with the highest levels of sample purity.

### Reverse Transcription and Amplification

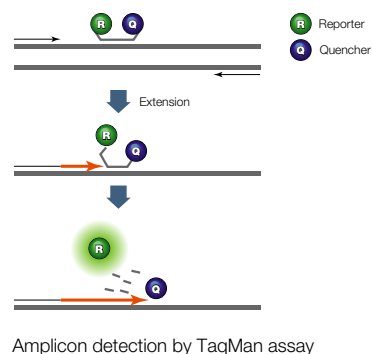
Bio-Rad offers preoptimized reagents for real-time PCR applications, all of which are specially formulated to achieve unrivaled results in real-time PCR. Bio-Rad's iScript™ reverse-transcription products are uniquely designed for sensitive and streamlined synthesis of cDNA. Real-time PCR supermixes based on the hot-start iTaq™ DNA polymerase offer sensitive and specific performance across a broad dynamic range with cDNA, plasmid DNA, and genomic DNA. Versatile enough for use with any fluorescent detection chemistry, real-time PCR reagents from Bio-Rad will accelerate your research.

### Sample Quality Assessment

The Experion™ automated electrophoresis system quickly delivers high-quality results for quantitation and integrity assessment of RNA, DNA, and protein samples. Experience meets innovation in each component of the Experion system, where the priming, vortex, and electrophoresis stations, as well as the RNA, DNA, and protein analysis kits, feature sleek, efficient designs that reliably provide accurate measurements of RNA or protein quantity and quality.

### Microarray Studies

The space-saving design of the BioOdyssey™ Calligrapher™ miniarrayer brings flexibility and ease of use directly to the benchtop for microarray printing applications. With minimal programming effort, the miniarrayer will print oligonucleotide, protein, or cell lysates onto glass slides, membranes, or 96-well plates. The BioOdyssey Calligrapher system is the perfect personal printing station for microarray applications.



Aurum RNA isolation kits



Reagents for real-time PCR



Experion automated electrophoresis system



BioOdyssey Calligrapher miniarrayer

## Ordering Information

Catalog # Description

### Thermal Cyclers and Real-Time PCR Detection Systems

|              |                                                                                                                                                                                                                                                                                                                        |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CFB-3120     | <b>MiniOpticon Real-Time PCR System</b> , includes optical housing, MJ Mini thermal cycler, analysis software                                                                                                                                                                                                          |
| 170-9780     | <b>iQ5 Multicolor Real-Time PCR Detection System</b> , includes iCycler base, optics module, software CD-ROM, 5 installed filter sets, 96-well reaction module, calibration solutions, optical-quality 96-well PCR plates, Microseal® 'B' seals, communication cables, power cord, quick reference cards, instructions |
| 170-9770     | <b>MyiQ Single-Color Real-Time PCR Detection System</b> , includes iCycler base, optics module, software CD-ROM, 96-well optical reaction module, optical-quality 96-well PCR plates, Microseal 'B' seals, communication cables, power cords, instructions                                                             |
| CFB-3260G    | <b>Chromo4 Four-Color Real-Time PCR Detection System</b> , includes optical housing, photonics shuttle, DNA Engine thermal cycler, 96-well sample block, analysis software                                                                                                                                             |
| CFB-3240G    | <b>Chromo4 Four-Color Real-Time PCR Detector</b> , includes optical housing, photonics shuttle, 96-well sample block, analysis software (complete system requires PTC-0200G or PTC-0221G)                                                                                                                              |
| PTC-0221G    | <b>Dyad Disciple Dual-Bay Thermal Cycler Chassis</b> , does not include Alpha units (requires 2)                                                                                                                                                                                                                       |
| 170-9753SE01 | <b>iQ5 Optical System Software</b> , Security Edition, single-seat license                                                                                                                                                                                                                                             |

### Accessories for iQ5 and MyiQ Systems

|          |                                                                                 |
|----------|---------------------------------------------------------------------------------|
| 223-9441 | <b>iQ™ 96-Well PCR Plates</b> , 25                                              |
| MSB-1001 | <b>Microseal 'B' Adhesive Seals</b> , 100                                       |
| TBS-0201 | <b>0.2 ml 8-Tube Strips Without Caps</b> , natural, 125 strips (1,000 tubes)    |
| TCS-0803 | <b>Optical Flat 8-Cap Strips</b> , for 0.2 ml tubes and plates, ultraclear, 120 |

### Accessories for Chromo4 and MiniOpticon Systems

|          |                                                                                                                                   |
|----------|-----------------------------------------------------------------------------------------------------------------------------------|
| HSP-9601 | <b>Hard-Shell Thin-Wall 96-Well Skirted PCR Plates</b> , white shell/clear well, 50                                               |
| HSP-9655 | <b>Hard-Shell Thin-Wall 96-Well Skirted PCR Plates</b> , white shell/white well, 50                                               |
| MSO-1001 | <b>Optical Film Sealing Kit</b> , for 96-well plates, includes optical compression pad, 100 Microseal 'B' clear adhesive seals    |
| MLL-4801 | <b>Multiplate™ Low-Profile 48-Well Unskirted PCR Plates</b> , 8 x 6 format, natural color, thin-wall polypropylene, package of 50 |
| MLL-4851 | <b>Multiplate Low-Profile 48-Well Unskirted PCR Plates</b> , white, 50                                                            |
| TLS-0801 | <b>Low-Profile 0.2 ml 8-Tube Strips Without Caps</b> , natural, 120 strips (960 tubes)                                            |
| TLS-0851 | <b>Low-Profile 0.2 ml 8-Tube Strips Without Caps</b> , white, 120 strips (960 tubes)                                              |
| TCS-0803 | <b>Optical Flat 8-Cap Strips</b> , for 0.2 ml tubes and plates, ultraclear, 120                                                   |

Catalog # Description

### Real-Time Reagents

|          |                                                                                                                                                                                                                                                            |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 172-5848 | <b>iQ Multiplex Powermix</b> , 50 x 50 µl reactions, 2x mix contains dNTPs, 11 mM MgCl <sub>2</sub> , iTaq DNA polymerase, stabilizers                                                                                                                     |
| 172-5849 | <b>iQ Multiplex Powermix</b> , 200 x 50 µl reactions                                                                                                                                                                                                       |
| 170-8880 | <b>iQ™ SYBR® Green Supermix</b> , 100 x 50 µl reactions, 2x mix contains 100 mM KCl, 40 mM Tris-HCl, pH 8.4, 0.4 mM each dNTP (dATP, dCTP, dGTP, dTTP), 50 U/ml iTaq DNA polymerase, 6 mM MgCl <sub>2</sub> , SYBR Green I, 20 nM fluorescein, stabilizers |
| 170-8882 | <b>iQ SYBR Green Supermix</b> , 500 x 50 µl reactions                                                                                                                                                                                                      |
| 170-8884 | <b>iQ SYBR Green Supermix</b> , 1,000 x 50 µl reactions                                                                                                                                                                                                    |
| 170-8860 | <b>iQ Supermix</b> , 100 x 50 µl reactions, 2x mix contains 100 mM KCl, 40 mM Tris-HCl, pH 8.4, 0.4 mM each dNTP (dATP, dCTP, dGTP, dTTP), 50 U/ml iTaq DNA polymerase, 6 mM MgCl <sub>2</sub> , stabilizers                                               |
| 170-8862 | <b>iQ Supermix</b> , 500 x 50 µl reactions                                                                                                                                                                                                                 |
| 170-8864 | <b>iQ Supermix</b> , 1,000 x 50 µl reactions                                                                                                                                                                                                               |

### Additional Literature

|      |                                                                                              |
|------|----------------------------------------------------------------------------------------------|
| 5215 | <b>Chromo4 Four-Color Real-Time PCR System Brochure</b>                                      |
| 5274 | <b>MiniOpticon Two-Color Real-Time PCR System Brochure</b>                                   |
| 5462 | <b>MyiQ and iQ5 Real-Time PCR Detection Systems Brochure</b>                                 |
| 5593 | <b>iQ5 Optical System Software, Security Edition (Version 2.0) Product Information Sheet</b> |

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Purchase of this instrument conveys a limited non-transferable immunity from suit for the purchaser's own internal research and development and for use in applied fields other than Human In Vitro Diagnostics under one or more of U.S. Patents Nos. 5,656,493, 5,333,675, 5,475,610 (claims 1, 44, 158, 160-163 and 167 only), and 6,703,236 (claims 1-7 only), or corresponding claims in their non-U.S. counterparts, owned by Applied Biosystems Corporation. No right is conveyed expressly, by implication or by estoppel under any other patent claim, such as claims to apparatus, reagents, kits, or methods such as 5' nuclease methods. Further information on purchasing licenses may be obtained by contacting the Director of Licensing, Applied Biosystems, 850 Lincoln Centre Drive, Foster City, California 94404, USA.

Bio-Rad's real-time thermal cyclers are licensed real-time thermal cyclers under Applied's United States Patent No. 6,814,934 B1 for use in research and for all other fields except the fields of human diagnostics and veterinary diagnostics.

Practice of the patented 5' Nuclease Process requires a license from Applied Biosystems. The purchase of these products includes an immunity from suit under patents specified in the product insert to use only the amount purchased for the purchaser's own internal research when used with the separate purchase of Licensed Probe. No other patent rights are conveyed expressly, by implication, or by estoppel. Further information on purchasing licenses may be obtained from the Director of Licensing, Applied Biosystems, 850 Lincoln Centre Drive, Foster City, California 94404, USA.

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Life Science  
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